

The Kryptos Moscow Billboard Problem

Hypothesizing an approach to decrypt K4 (October 18, 2019)

*"How do you put Kryptos on a billboard in Moscow, so it means something?" – 2011
DC Gather (Elonka Dunin 2011 dinner notes)*

The literal interpretation is putting the 7 characters "KRYPTOS" on a billboard in Moscow. Cryptically speaking, Mr. Scheidt has revealed several clues on approaching K4 after the billboard problem is fundamentally understood. Imagine during the Cold War an encoded message from Langley has to be sent to a Moscow operative in deep cover. The only way to send the message is through a public billboard that can be seen and analyzed by the Russian government. Armed with only pencil and paper, what is the minimum decryption steps required to thwart Russian computational brute force yet have the operative execute an analog decryption?

To simplify the problem, some assumptions are made:

1. The message will be KRYPTOS (English characters only for simplicity) in a single billboard
2. Russian intelligence can also see the billboard and have unlimited computing power
3. Langley will deliver the message in the future and there is no contact between the operative and headquarters (Langley) implying only basic cryptographic skills for substitution and transposition required
4. The operative can only decrypt with pencil and paper
5. Ciphertext is vulnerable to brute force when
 - a. Single substitution step OR
 - b. Single transposition step

The questions to be answered:

- i. What is the minimum number of decoding steps (substitution/transposition) to thwart brute forcing?
- ii. What is the minimum number of keys that must also be distributed on the billboard?
AND
- iii. If the operative was compromised, what additional countermeasures could be employed (i.e. masking)?

Important terms:

Substitution ciphers – where a plaintext character is replaced by another character and the position of the character is preserved

Conventional substitution – utilizes a standard Vigenère tabula recta for encryption/decryption requiring only a key

Unconventional substitution – the decoding tableau (tabula recta) is altered requiring both key(s) and the modified tableau

Transposition ciphers – where all plaintext characters are unchanged BUT may change position; a key is optional

Key – the fundamental component used in a cryptographic algorithm that may transform plaintext into ciphertext or ciphertext into plaintext

Masking – a destructive and irreversible encryption process that removes characters

Summary of Decryption Steps

	Decryption sequence	Number of keys and/or deciphering tableau	Vulnerable to brute force
Single Step	Substitution	Conventional polyalphabetic: 1 key Unconventional polyalphabetic: 1 key plus tableau	Yes to all
	Transposition	0 if self-explanatory, or 1+ if more difficult	Yes due to plaintext being scrambled
Double Step	Double substitution	Conventional polyalphabetic: 2 keys Unconventional polyalphabetic: 2 keys plus tableau	Yes due to fundamentally being a single substitution in aggregate
	Double transposition	2+ keys if multiple algorithms	Yes due to fundamentally being a single transposition in aggregate
	Substitution then transposition*	Conventional polyalphabetic: 1 key (special case) or 2 keys Unconventional polyalphabetic: 2 keys plus tableau	Yes* for special case No
	Transposition then substitution*	Conventional polyalphabetic: 1 key (special case) or 2 keys Unconventional polyalphabetic: 2 keys plus tableau	No

Special Case Double Decryption Steps

Substitution then transposition (1 key) – the key would both provide the substitution and transposition algorithm (i.e. 7-5-7-5-7 could indicate odd characters shift 7 and even characters shift 5 and then rearrange to a matrix of 7-5-7-5-7). If the Russians were able to infer either substitution or transposition from the key, then the corresponding substitution/transposition could be brute forced.

Transposition then substitution (1 key) – this is a special case scenario where the ciphertext is also the substitution (conventional) key after transposition. The plaintext is recovered after the transformed ciphertext is used to decode the original ciphertext with conventional substitution decryption methods (XOR, Vernam, OTP, Vigenère tabula recta, running key, etc.)

Ciphertext: KRYPTOS
 Transposed ciphertext: RKYOTPS (conventional substitution decrypt)
 Decoded plaintext: THABAZA

This could possibly result in “The Baza (cuckoo-hawk)” as a codename. From the table above, the decryption steps that would best thwart brute forcing and has the minimum number of keys is the special case scenario of transposition then substitution to decode.

Masking

Unlike substitution ciphers which replace one character for another, masking is an irreversible process that removes characters without fundamentally changing the position of the plaintext. The destructive nature would also imply that this would be done at the plaintext level. For encryption, this would be done first, and for decryption it would end up as some form of concealment as some English character statistics would be altered. If it were applied to the ciphertext or intermediaries, it may compromise the integrity of the ciphers and plaintext in its totality.

Codenames (Nicknames)

In addition to masking, having a codename would also provide an extra layer of protection if it was referring to the name of a target or asset. For example, TRIGON refers to Aleksandr Ogorodnik during the Cold War.

Conclusion

The crux of the Moscow Billboard problem lies with the number of encryption steps and the quantity of keys that also need to be delivered. When the quantity of ciphers is increased, the number of keys may also increase. With a single billboard, it becomes increasingly difficult to include the keys and/or deciphering tableau without garnering the attention of Russian Intelligence.

If the analysis of Moscow Billboard is analogous to the K4 section of Kryptos, then the encoding of K4 would be masking, substitution, then transposition. Conversely the decrypting steps of K4 would then involve:

1. Transposition of ciphertext to produce the substitution key
2. Substitution decryption using transposed ciphertext
3. Analysis of masked plaintext

Generally speaking, the method being described is a form of asymmetrical public key cryptography where the user has to reassemble the decryption key from transforming the ciphertext to then apply a substitution decryption process.

To decode manually, this would require a circular process of transforming K4 ciphertext then decrypting K4 from the transformed substitution key. Fortunately, Jim Sanborn did offer two words for a partial crib: BERLIN and CLOCK. If the Moscow Billboard steps are used the K4 ciphertext characters NYPVTTMZFPK can be decrypted with MUYKLGKORNA using a standard polyalphabetic substitution (XOR, Vernam, OTP, Vigenère Tabula Recta, running key, etc.).